Project Proposal
Cohort A
Group 7

Version:1.0

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1. Project Summary

1.1 Purpose

This proposal describes the plan for the Outvier EdTech project. outlines the plan for the Outvier EdTech Project. Our primary goal is to obtain approval from to proceed with the development of a Benchmarking Dashboard, a web application designed to help international students make informed college selection decisions. secure approval to proceed with the development of a Comparative Analytics Dashboard—a web-based application designed to assist international students in making informed university selection decisions. By leveraging modern web technologies and artificial intelligence, this project aims to transform fragmented public data into actionable insights. transform fragmented public data into actionable insights.

1.2 Business and Policy Context

In today's global educational landscape, international students face significant obstacles when planning their studies. the current global education landscape, international students face significant barriers when planning their studies. Relevant data, such as tuition fees, visa pathways and career outcomes, is often scattered across hundreds of university websites, ranking portals and government databases. data—such as tuition fees, visa pathways, and career outcomes—is often scattered across hundreds of university websites, ranking portals, and government databases. This fragmentation forces students to manually compile information, a process that is both time-consuming and error-prone. prone to error.

The Outvier project aligns with the broader strategic objective of improving accessibility in higher education. goal of enhancing accessibility in higher education. By standardizing this information, we support institutional transparency and enable students to make choices better suited to their financial and career aspirations. empower students to make choices that better align with their financial and career aspirations.

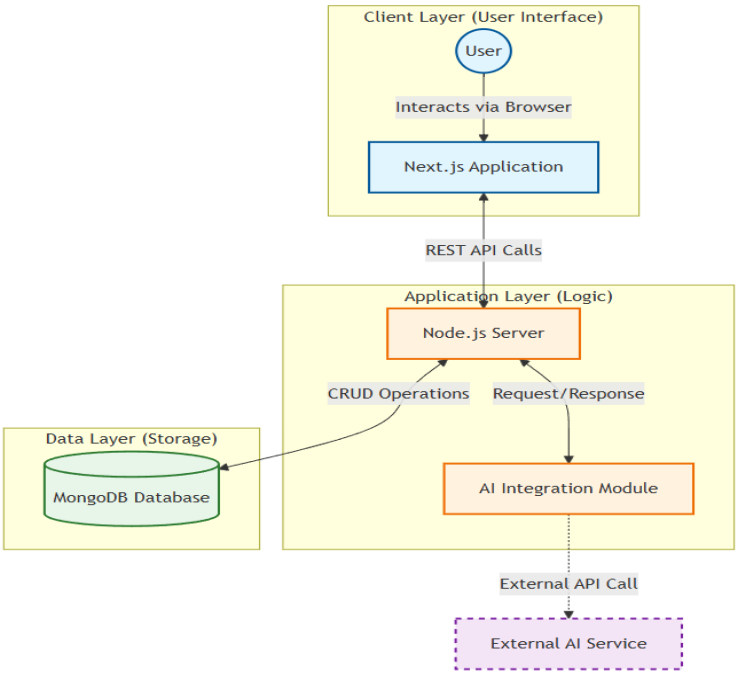

Figure 1 High-Level System Architecture

1.3 Working Title for the Proposed Project

The working title of the project is the Outvier EdTech Dashboard Project. for the project is the Outvier EdTech Dashboard Project.

1.4 Objective(s)

The main objective of the Outvier project is to develop a centralized platform that simplifies the university selection process through data visualization and intelligent comparison. primary objective of the Outvier Project is to develop a centralized platform that simplifies the university selection process through data visualization and intelligent comparison. Specifically, our goal is to:

- Aggregate and standardize disparate academic data into a unified, searchable format.
- Implement advanced filtering and ranking logic to allow students to compare institutions on multiple dimensions (e.g. across multiple dimensions (e. g. , cost compared vs. classification). ranking).
- Integrate AI-based recommendations to provide personalized program suggestions based on user profiles. AI-driven recommendations to provide personalized program suggestions based on user profiles.

1.5 Project Complexity

We rate this project as having moderate to high complexity. assess this project as having moderate to high complexity. This assessment is based on the need to integrate a modern frontend framework (Next. js) with a scalable backend (Node. js), while implementing complex data visualization libraries (D3. also implementing complex data visualization libraries (D3. js). Additionally, integrating external AI APIs for recommendation capabilities adds a layer of technical dependency that requires careful architectural planning. Furthermore, the integration of external AI APIs for recommendation features adds a layer of technical dependency that requires careful architectural planning.

1.6 Potential Benefits

- Decision-making efficiency: Reduce the time students spend researching programs from weeks to hours. Successful implementation of the Outvier dashboard will offer several key benefits:
- Decision Efficiency: Reducing the time students spend researching programs from weeks to hours.
- Data Accuracy: provide standardized comparisons (e.g. Providing standardized comparisons (e. g. , unified currency conversion) which reduce the risk of financial calculation errors. that reduce the risk of financial miscalculations.

Strategic Overview: Providing users with a visual understanding of the trade-offs between university rankings, costs and employment outcomes. Insight: Offering users a visual understanding of the trade-offs between university rankings, costs, and employment outcomes.

1.7 Feasibility Statement

The project is deemed feasible given the team's existing skills in full-stack development and the availability of open source tools. skill set in full-stack development and the availability of open-source tools. The main constraint is data acquisition; however, this will be mitigated by using a defined dataset of representative universities for the prototype. primary constraint is data acquisition; however, this will be mitigated by using a defined dataset of representative universities for the prototype. The team has access to the IT resources and development environments necessary for delivery. computational resources and development environments required for delivery.

1.8 Recommendation

It is recommended that the project proceed as indicated. outlined. The proposed solution addresses a real user problem and leverages a technology stack that is both robust and adapted to the team's capabilities. genuine user pain point and leverages a technology stack that is both robust and appropriate for the team's capabilities.

2. Business Assessment

2.1 Situation Assessment and Problem Statement

Currently, prospective international students do not have a unified tool to compare universities based on holistic parameters. lack a unified tool to compare universities across holistic metrics. Although ranking sites exist, they often fail to provide precise details on specific courses or "informal" factors such as visa pathways and return on investment (ROI).

The Gap: There is a disconnect between the vast amount of public data and the user's ability to synthesize that data to make a confident decision. into a confident decision. Students often experience “analysis paralysis.” "analysis paralysis. “The solution: Outvier fills this gap by acting as an analytics layer on top of raw data. "

The Solution: Outvier bridges this gap by acting as an analytics layer over raw data. Instead of simply listing universities, the system visualizes the relationships between them, showing, for example, which universities offer the best employment rates for a specific educational bracket. just listing universities, the system visualizes the relationships between them—showing, for example, which universities offer the best employment rates for a specific tuition bracket.

2.2 Options considered

Three options were evaluated during the initial planning phase:

- Option 1 - **Do Nothing**: Students continue to use multiple websites and worksheets. spreadsheets. This does not resolve the identified issue.
- Option 2 - **Static Information Portal**: A basic website displaying static university profiles. This is easier to build but fails to achieve the goal of "benchmarking" and dynamic filtering.
- Option 3 - **The Preferred Option**: A dynamic web application with a database backend, visualization tools and AI integration. tools, and AI integration. This is the most resource-intensive option, but it is the only one that fully meets the objectives and complexity requirements of the project.

2.3 Consultation

Initial discussions took place with the project supervisor and potential student users. have been held with the project supervisor and potential student users. Feedback indicated strong demand for “side-by-side” comparison features and “estimated ROI” calculations. a high demand for "side-by-side" comparison features and "estimated ROI" calculations. There is strong support for this concept, particularly regarding the inclusion of career outcomes data. concept, particularly regarding the inclusion of career outcome data.

3. Proposed Scope

3.1 Scope Definition

Table 1 Project Scope Elements

Element	Detail
Objective	Provide a working prototype of a benchmarking dashboard. To deliver a working prototype of a Comparative Analytics Dashboard.
Outcome	A user-friendly web application where students can filter, compare and visualize academic data. compare, and visualize university data
Output	1. Functional Source Code (Frontend & Backend). 2. MongoDB Database with standardized sample data. 3. Technical Documentation (User Manual, Test Plan).
Quality Citeria	1. The dashboard must render visualizations accurately using D3. js. 2. Data retrieval latency should be less than 2 seconds for standard queries. must be under 2 seconds for standard queries. 3. The app should be responsive on all desktop devices. application must be responsive across desktop devices.
Customer(s)	The main clients are international students seeking higher education opportunities.

3.2 Assumptions

To proceed with planning, the following assumptions have been made:

- Fees and rankings can be legally collected and used for educational, non-commercial purposes.
- The team members have a baseline proficiency in JavaScript (React/Node) and will upskill in D3. js and MongoDB during the initial development phase.
- The AI API service used for recommendations will have a free tier sufficient for prototype traffic.

3.3 Constraints

- Deadline: The project must be delivered within a semi-annual deadline of 12 weeks.
- Budget: The project should use open source frameworks and free cloud services to maintain zero cost.
- Data: The prototype will be limited to a dataset of approximately 50 universities and 200 courses to ensure data quality and speed of development. development speed.

3.4 Scope of Work

In Scope:

- Development of the frontend user interface under Next.
- Creating backend REST API using Node. Creation of backend REST APIs using Node. js/Express.
- Design and implementation of databases in MongoDB. Database design and implementation in MongoDB.
- Integration of D3. js for visual analysis (charts, charts). analytics (charts, graphs).
- Implementation of an AI-based recommendation functionality.

Out of Scope:

- Development of Native mobile applications (iOS/Android). application development (iOS/Android).
- Live, real-time data extraction from academic websites (static dataset only). scraping from university websites (static dataset only).
- Processing User payments or managing subscriptions. payment processing or subscription management.

3.5 Implementation Strategy

The project will follow an Agile-inspired methodology, broken down into two-week sprints. This allows for iterative feedback from the supervisor and ensures that core functionality is prioritized.

Table 2 Initial project plan

Element	Detail	Issues
Project Schedule	The project is expected to commence on March, 30 and be completed by June, 01. Key milestones include: - Week 4: Proposal Submission - Week 8: Specification & Design Document - Week 11: Prototype Completion - Week 13: Final Presentation	Estimates assume constant team availability. Estimations assume consistentteam availability. A 1 week buffer is included for testing. of 1 week is included for testing.

Budget Estimate	Total Budget: $0. Use of student laptops, free cloud hosting (Vercel/Render), free online API and academic software licenses provided by VIT. Utilizing student laptops, free-tier cloud hosting (Vercel/Render), free-online api andacademic software licenses provided by VIT.	No financial risk involved. All tools are open source or free. open-source or free-tier.
Other Resources	- 5x Development Laptops - GitHub Repository (Version Control) - VS Code (IDE) - Access to VIT Library for research.	Reliance on personal internet connections for remote work.

4. Project Management Outline

4.1 Governance

The project governance structure consists of:

- Project Sponsor: [Supervisor Name], VIT Academic Supervisor.
- Project Manager: [Student Name], Group Leader.
- Development Team: [Remaining 4 Student Names].

4.2 Key Risks and Issues

We have identified the following main risks:

1. **Technical Learning Curve:** D3.js has a steep learning curve.
   - *Mitigation*: Assigning a dedicated team member to research D3.js early and conduct knowledge-sharing sessions.
2. **Data Inconsistency:** Public data may be outdated or formatted differently across sources.
   - *Mitigation*: Defining a strict data schema and manually verifying a subset of data to ensure the comparison logic works correctly.
3. **Integration Issues:** Connecting the frontend to the AI API may present latency or formatting challenges.
   - *Mitigation*: developing a simple "proof of concept" integration in the early stages to validate the architecture.

5. References

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